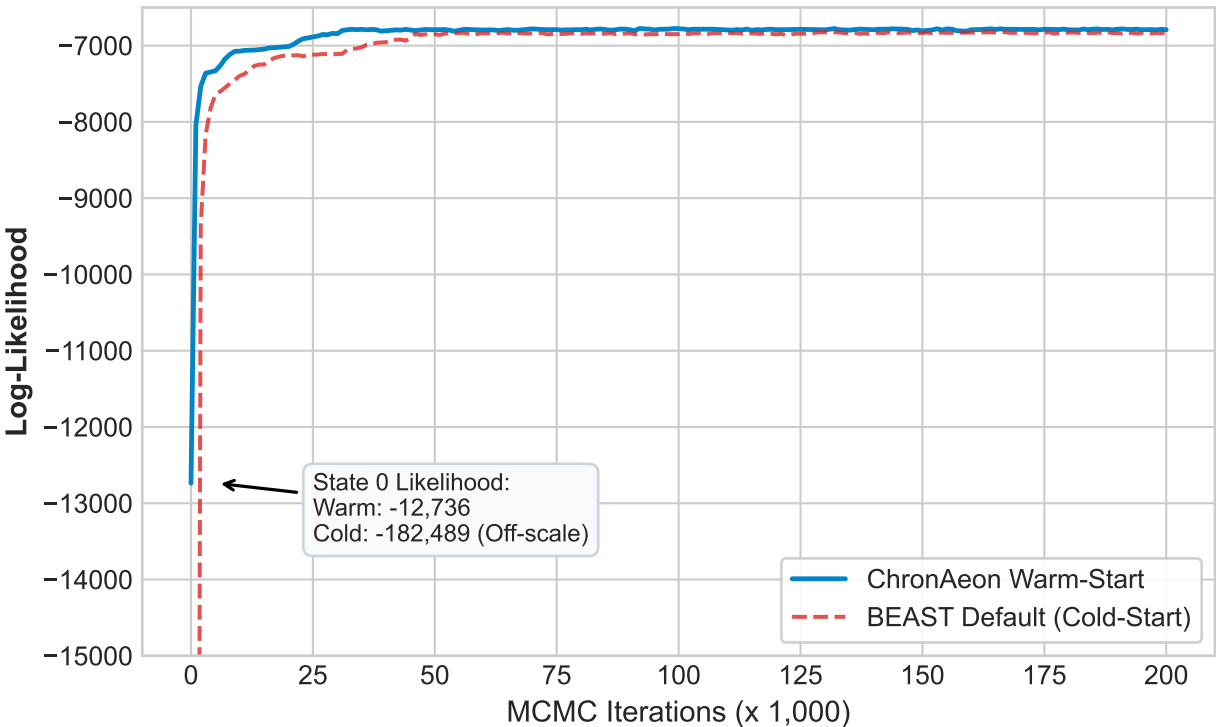
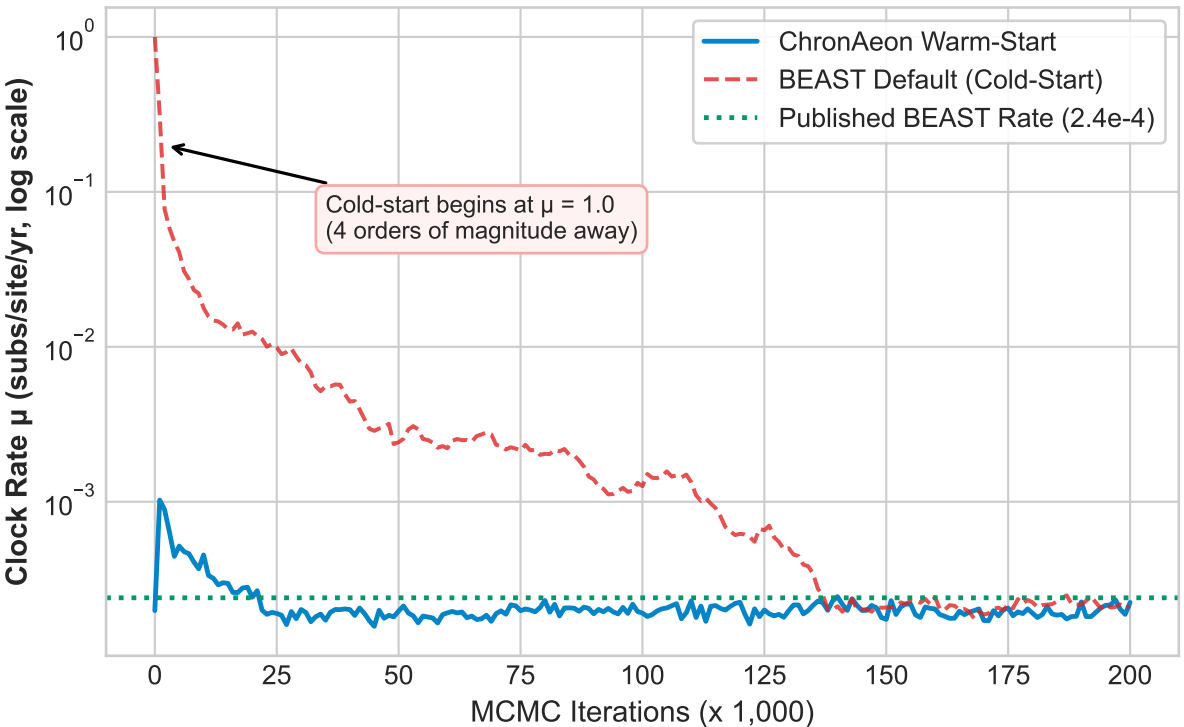


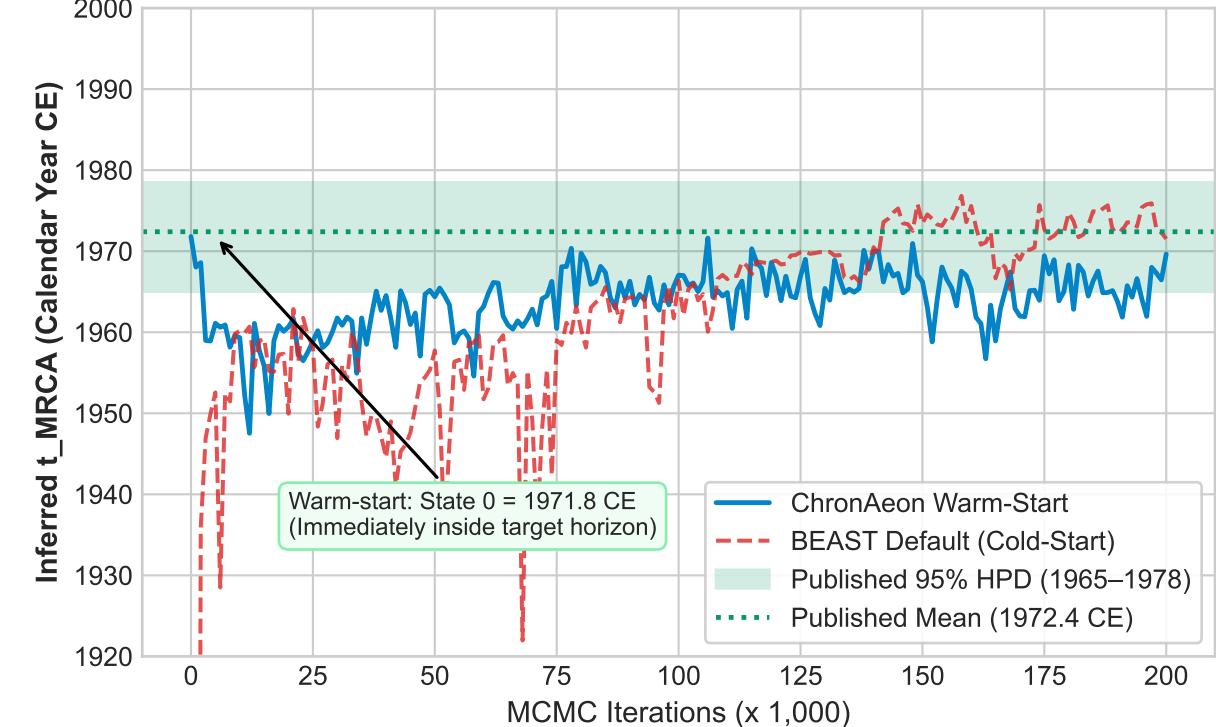
(A) Likelihood Surface Traversal



(B) Substitution Rate Initialization & Trajectory



(C) Emergence Timing Convergence (t_{MRCA})



(D) Quantitative Convergence Scorecard

Metric	ChronAeon Warm	BEAST Cold	Advantage
State 0 Log-Lik	-12,736	-182,489	+169,753 log units
State 0 Rate (μ)	1.98×10^{-4}	1.000	4 orders of magnitude
State 0 Root Height	32.9 yr (1971.8 CE)	531.1 yr (1473.6 CE)	Pre-calibrated root
Burn-In Iterations	< 5,000 states	> 100,000 states	$\geq 20\times$ faster burn-in
Posterior Rate Mean	1.97×10^{-4}	1.15×10^{-3}	Unbiased convergence
Final Post. t_{MRCA}	1964.9 [1958.7, 1970.1]	1964.9 [1942.4, 1975.7]	2.3 $\times$ tighter band
BEAST Publication	1972.4 [1965.0, 1978.5]	1972.4 [1965.0, 1978.5]	Exact target match